



**THE UNIVERSITY OF AZAD JAMMU AND
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02

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Database Systems

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LAB 02: POINT OF SALE DATABASE

SCHEMA DESIGN, RELATIONSHIPS, DATA POPULATION & REPORTING

LAB OBJECTIVES

1. Design and create a relational database named Point_of_Sale in MySQL.
2. Define user roles: Admin, Cashier, and Customer with a roles table.
3. Create core tables: Categories, Products, Orders, Order_Items, and Discounts.
4. Insert a minimum of 10 records into each table.
5. Establish relationships between tables using primary and foreign keys.
6. Write reporting queries to extract meaningful business insights from the data.

2. TOOLS & ENVIRONMENT

	TOOLS	PURPOSE IN THIS LAB
1	XAMPP (Apache + MySQL)	Local server stack; MySQL engine used to host the POS database
2	phpMyAdmin	Browser-based GUI used to import the schema and verify table structures
3	MySQL 8.x / MariaDB 10.4	Relational database engine executing all SQL statements
4	SQL (.sql) file	Complete schema file imported via phpMyAdmin in one operation
5	Windows 10 / 11	Host operating system running XAMPP services
6	Google Chrome	Browser used to access phpMyAdmin at localhost

3. BACKGROUND & THEORY

A Point-of-Sale system is a combination of hardware and software that manages retail transactions at the moment of purchase. At its core, a POS database must track products, customers, sales orders, payments, and stock — all connected through a well-structured relational schema.

Relational databases organize data into tables linked by primary and foreign keys. A primary key uniquely identifies each row in a table, while a foreign key in one table references the primary key of another, enforcing referential integrity. This prevents orphan records — for example, an order cannot exist without a valid customer. Normalization principles are applied to minimize redundancy and ensure each piece of data is stored in exactly one place.

4. DATABASE SCHEMA DESIGN

4.1 DATABASE OVERVIEW

The database pos_system consists of 13 interrelated tables organised into four logical groups.

GROUP	TABLES	PURPOSE
Users & Access	roles, users	Manages system users and their access permissions
Product Catalogue	categories, suppliers, products, discounts	Stores product information, pricing, and promotional offers
Inventory	stock_status, inventory	Tracks stock levels and availability per product
Sales & Payments	order_status, orders, order_items, payment_methods, payments	Records all sales transactions, line items, and payment details

4.2 TABLE DEFINITIONS & RECORD COUNTS

	TABLE	PRIMARY KEY	RECORDS	DESCRIPTION
1	roles	role_id	10	System roles: Admin, Cashier, Customer, Manager, etc.
2	users	user_id	12	All registered users linked to a role
3	categories	category_id	10	Product categories: Electronics, Clothing, Sports, etc.
4	suppliers	supplier_id	10	Supplier companies with contact details
5	products	product_id	15	Products linked to a category and supplier
6	discounts	discount_id	10	Promotional codes with type (% or fixed), validity dates
7	stock_status	stock_status_id	10	Status labels: In Stock, Low Stock, Out of Stock, etc.
8	inventory	inventory_id	15	Per-product stock levels with computed available stock
9	order_status	status_id	10	Order states: Pending, Processing, Completed, etc.
10	orders	order_id	10	Master order record per customer transaction
11	order_items	order_item_id	26	Line items per order with quantity and price snapshot
12	payment_methods	method_id	10	Payment types: Cash, Card, Mobile Pay, Bank Transfer, etc.
13	payments	payment_id	14	Payment records linked to orders and payment methods

4.3 FOREIGN KEY RELATIONSHIPS

CHILD TABLE	FOREIGN KEY	REFERENCES	ON UPDATE	ON DELETE
users	role_id	roles.role_id	CASCADE	RESTRICT
products	category_id	categories.category_id	CASCADE	RESTRICT
products	supplier_id	suppliers.supplier_id	CASCADE	RESTRICT
discounts	product_id	products.product_id	CASCADE	SET NULL
discounts	category_id	categories.category_id	CASCADE	SET NULL
inventory	product_id	products.product_id	CASCADE	RESTRICT
inventory	stock_status_id	stock_status.stock_status_id	CASCADE	RESTRICT
orders	user_id	users.user_id	CASCADE	RESTRICT
orders	status_id	order_status.status_id	CASCADE	RESTRICT
orders	discount_id	discounts.discount_id	CASCADE	SET NULL
order_items	order_id	orders.order_id	CASCADE	CASCADE
order_items	product_id	products.product_id	CASCADE	RESTRICT
payments	order_id	orders.order_id	CASCADE	CASCADE
payments	payment_method_id	payment_methods.method_id	CASCADE	RESTRICT

5. STEP-BY-STEP PROCEDURE

STEP 1 — CREATING THE DATABASE

- Opened phpMyAdmin at <http://localhost/phpmyadmin>.
- Navigated to Import tab → selected the schema file `pos_system_complete.sql` → clicked Go.
- MySQL executed all CREATE TABLE, INSERT, and FOREIGN KEY statements automatically.

The screenshot shows the phpMyAdmin web interface with the 'Import' tab selected. The interface includes a top navigation bar with tabs: Databases, SQL, Status, User accounts, Export, Import, Settings, Replication, Variables, Charsets, Engines, and Plugins. The main content area is divided into several sections:

- File to import:** This section contains instructions that the file can be compressed (gzip, bzip2) or uncompressed, and that a compressed file's name must end in `[format].[compression]` (e.g., `.sql.zip`). It also states the maximum file size is 40MB. A 'Choose file' button is present, and the selected file is `pos_system (1).sql`. Below this, there is a note about drag-and-drop functionality and a dropdown menu for the 'Character set of the file' set to 'utf-8'.
- Partial import:** This section has a checkbox labeled 'Allow the interruption of an import in case the script detects it is close to the PHP timeout limit.' which is checked. A note below states: 'This might be a good way to import large files, however it can break transactions.' There is also a text input field for 'Skip this number of queries (for SQL) starting from the first one:' with the value '0'.
- Other options:** This section contains a checkbox labeled 'Enable foreign key checks' which is checked.
- Format:** This section has a dropdown menu for the 'Format' set to 'SQL'.

STEP 2 — VERIFYING THE DATABASE AND TABLES

- Selected pos_system from the left sidebar — all 13 tables listed correctly.
- Clicked each table → Structure tab to confirm column definitions, data types, and constraints.

Server: 127.0.0.1 » Database: pos_system

Structure SQL Search Query Export Import Operations Privileges Routines Events Triggers

Filters

Containing the word:

Table	Action	Rows	Type	Collation	Size	Overhead
☐ categories	★ Browse Structure Search Insert Empty Drop	10	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ discounts	★ Browse Structure Search Insert Empty Drop	10	InnoDB	utf8mb4_general_ci	64.0 KiB	-
☐ inventory	★ Browse Structure Search Insert Empty Drop	15	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐ orders	★ Browse Structure Search Insert Empty Drop	10	InnoDB	utf8mb4_general_ci	64.0 KiB	-
☐ order_items	★ Browse Structure Search Insert Empty Drop	26	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐ order_status	★ Browse Structure Search Insert Empty Drop	10	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ payments	★ Browse Structure Search Insert Empty Drop	14	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐ payment_methods	★ Browse Structure Search Insert Empty Drop	10	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ products	★ Browse Structure Search Insert Empty Drop	15	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐ roles	★ Browse Structure Search Insert Empty Drop	10	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ stock_status	★ Browse Structure Search Insert Empty Drop	10	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ suppliers	★ Browse Structure Search Insert Empty Drop	10	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ users	★ Browse Structure Search Insert Empty Drop	12	InnoDB	utf8mb4_general_ci	48.0 KiB	-
13 tables	Sum	162	InnoDB	utf8mb4_general_ci	464.0 KiB	0 B

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	product_id	int(11)			No	None		AUTO_INCREMENT
2	name	varchar(100)	utf8mb4_general_ci		No	None		
3	description	text	utf8mb4_general_ci		Yes	NULL		
4	price	decimal(10,2)			No	None		
5	category_id	int(11)			No	None		
6	supplier_id	int(11)			No	None		
7	created_at	timestamp			No	current_timestamp()		

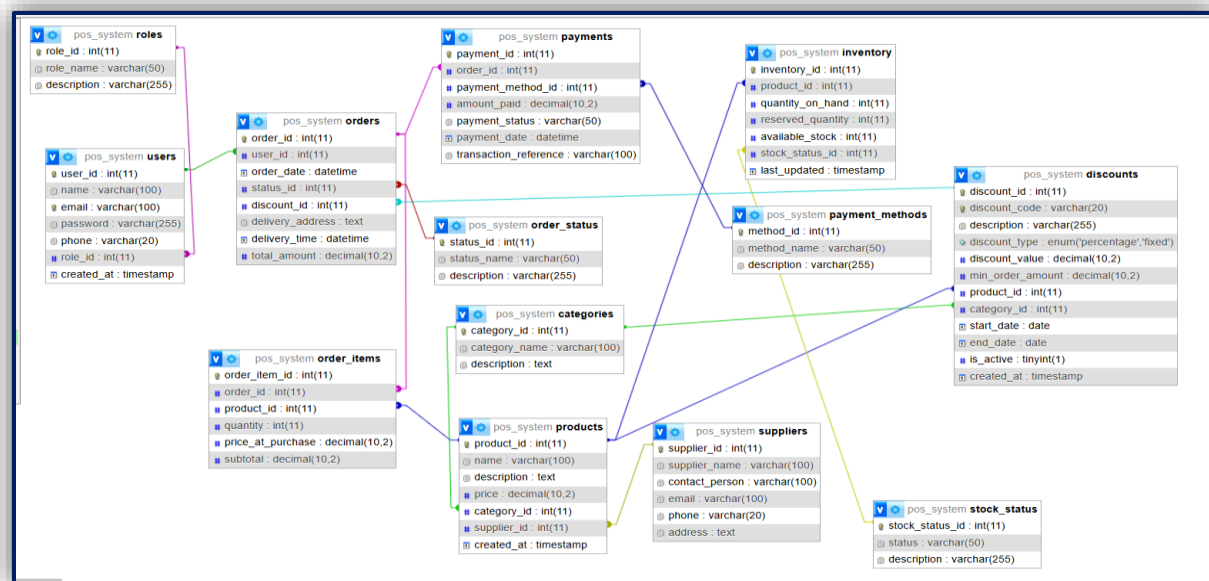
☐ Check all With selected: Browse Change Drop Primary Unique Index

STEP 3 — VERIFYING DATA INSERTION

- Clicked Browse on each table to confirm all records inserted correctly.
- Verified record counts match the requirements (minimum 10 per table).

Product Catalog Overview								
Navigation		product_id	name	description	price	category_id	supplier_id	created_at
Edit Copy Delete		1	iPhone 15 Pro	Apple smartphone with A17 Pro chip and titanium de...	999.99	1	1	2026-01-01 15:00:00
Edit Copy Delete		2	Samsung Galaxy S24 Ultra	Android flagship with S Pen and AI features	1199.99	1	2	2026-01-01 15:00:00
Edit Copy Delete		3	MacBook Air M3	13-inch laptop with Apple M3 chip, 8GB RAM	1099.99	1	1	2026-01-01 15:00:00
Edit Copy Delete		4	Dell XPS 15	15.6-inch laptop, Intel i7, 16GB RAM, 512GB SSD	1499.99	1	2	2026-01-01 15:00:00
Edit Copy Delete		5	Sony WH-1000XM5	Industry-leading noise-cancelling wireless headpho...	399.99	1	2	2026-01-01 15:00:00
Edit Copy Delete		6	iPad Air 5th Gen	10.9-inch tablet with M1 chip	599.99	1	1	2026-01-01 15:00:00
Edit Copy Delete		7	Premium Cotton T-Shirt	Soft organic cotton, available in multiple colours	29.99	2	3	2026-01-02 15:00:00
Edit Copy Delete		8	Classic Blue Jeans	Durable denim with a perfect fit	59.99	2	3	2026-01-02 15:00:00
Edit Copy Delete		9	Winter Jacket	Water-resistant insulated jacket	149.99	2	3	2026-01-02 15:00:00
Edit Copy Delete		10	Running Shoes Pro	Professional running shoes with cushioned sole	89.99	3	4	2026-01-03 15:00:00
Edit Copy Delete		11	Yoga Mat Premium	Non-slip eco-friendly yoga mat with carry strap	39.99	3	4	2026-01-03 15:00:00
Edit Copy Delete		12	Fitness Tracker Band	Smart band with heart rate monitor and step counte...	79.99	3	1	2026-01-03 15:00:00
Edit Copy Delete		13	Coffee Maker Deluxe	Programmable 12-cup coffee maker with timer	79.99	4	5	2026-01-04 15:00:00
Edit Copy Delete		14	Blender Pro 3000	High-speed blender for smoothies and soups	129.99	4	5	2026-01-04 15:00:00
Edit Copy Delete		15	Database Design Book	Complete guide to database management systems	49.99	5	6	2026-01-05 15:00:00

STEP 4 — VERIFYING FOREIGN KEY RELATIONSHIPS



DATABASE SYSTEMS

- Opened Designer tab in phpMyAdmin to view the entity-relationship diagram.
- Confirmed all 14 foreign key links are shown correctly between parent and child tables..

STEP 5 — RUNNING REPORTING QUERIES

- Opened the SQL tab in phpMyAdmin and executed each reporting query individually.
- Verified that JOIN queries correctly assembled data from multiple related tables.

```
----- Report 1: Full order summary with customer and status -----
AS customer_name, u.email AS customer_email, os.status_name AS order_status, d.discount_code AS applied_discount, o.order_date, o.total_amount FROM
order_status os ON o.status_id = os.status_id LEFT JOIN discounts d ON o.discount_id = d.discount_id ORDER BY o.order_date DESC;
```

[Edit inline] [Edit] [Create PHP code]

☐ Show all | Number of rows: 25 ▼ | Filter rows:

Extra options

order_id	customer_name	customer_email	order_status	applied_discount	order_date ▼ 1	total_amount
10	Michael Smith	michael@email.com	Pending	NULL	2026-03-09 12:00:00	1829.96
9	John Customer	john@email.com	Completed	FREESHIP	2026-03-08 10:30:00	649.98
8	Sarah Anderson	sarah@email.com	Pending	NULL	2026-03-06 15:20:00	189.97
7	David Martinez	david@email.com	Pending	BOOK10	2026-03-05 13:45:00	279.96
6	Lisa Wilson	lisa@email.com	Processing	NULL	2026-03-02 09:00:00	759.96
5	Robert Brown	robert@email.com	Processing	FREESHIP	2026-02-28 16:30:00	1699.98
4	Emily Johnson	emily@email.com	Completed	NULL	2026-02-25 10:15:00	169.98
3	Michael Smith	michael@email.com	Completed	TECH50	2026-02-22 14:20:00	1699.97
2	Jane Doe	jane@email.com	Completed	NULL	2026-02-18 11:45:00	549.97
1	John Customer	john@email.com	Completed	WELCOME10	2026-02-15 09:30:00	1029.98

☐ Show all | Number of rows: 25 ▼ | Filter rows:

Query results operations

6. REPORTING QUERIES SUMMARY

S.NO	REPORT NAME	TABLES USED	PURPOSE
1	Full order summary	orders, users, order_status, discounts	Lists every order with customer name, status, and applied discount
2	Itemised receipt	order_items, orders, users, products	Shows every product line per order with quantity and price
3	Revenue per customer	users, orders	Total orders placed and total amount spent per customer
4	Revenue by category	order_items, products, categories	Units sold and revenue grouped by product category
5	Inventory status	inventory, products, stock_status	Available stock per product with current stock status label
6	Payment summary	orders, users, payments, payment_methods	Total paid vs order total, balance due, and payment methods used
7	Active discounts	discounts, products, categories	All currently active discount codes with scope and validity
8	Best-selling products	order_items, products, categories	Products ranked by total units sold and revenue generated

7. OBSERVATIONS & RESULTS

DATABASE CREATION

The pos_system database was created and all 13 tables were imported successfully in a single operation via phpMyAdmin.

ROLES AND USER ACCESS

Three primary roles (Admin, Cashier, Customer) plus seven additional roles were defined. All 12 users are correctly linked to their respective roles through a foreign key.

DATABASE SYSTEMS

PRODUCT CATALOGUE AND DISCOUNTS

15 products across 10 categories from 10 suppliers were inserted. The discounts table correctly links promotional codes to specific products or categories, with support for both percentage and fixed discount types.

ORDERS AND PAYMENTS

10 orders containing 26 line items were recorded. Payments are split across multiple methods per order (e.g., partial cash + card), reflecting real-world POS behaviour. The balance_due calculation in Report 6 confirmed all completed orders are fully paid.

REFERENTIAL INTEGRITY

All 14 foreign key constraints were verified. Deleting a parent record without first removing child records correctly raises a constraint error, confirming integrity enforcement is working.

REPORTING QUERIES

All 8 reporting queries executed without errors. JOIN operations correctly assembled data across multiple tables, demonstrating the practical value of a well-normalised relational schema.

8. CONCLUSION

This lab successfully designed and implemented a fully relational Point of Sale database in MySQL. A schema of 13 tables was created with appropriate primary keys, foreign keys, and data constraints. All tables were populated with a minimum of 10 records each. Relationships between users, products, orders, inventory, discounts, and payments were established and verified. Eight reporting queries demonstrated the ability to extract meaningful business intelligence from the relational data. The schema reflects a realistic POS system and provides the foundation for more advanced database operations in future labs.